Lecture 9

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Midterm 1

- Reminder that Midterm 1 will be on Wednesday, October 14 in class.
- Midterm 1 will cover:
  - Chapters 1-5 and 8-12 in the text, and
  - the material in Lectures 1-8.
- The R material covered in lab will not be on Midterm 1 but is fair game for Midterm 2 and the final.

Midterm 1 (continued)

- You can find example questions for the exam from:
  - Sample midterms posted on Canvas
  - Homework questions
  - Additional questions in the text
  - Review questions
  - Questions discussed in lecture
- You will want to focus more on solving problems and less on memorizing the lectures.

Lecture 9

1. The Regression Effect and Regression Fallacy
2. Using Regression
3. Reading Summary – Chapters 13 and 14
4. Basic Concepts and Notation for Probability
5. Bounds on Joint Probabilities

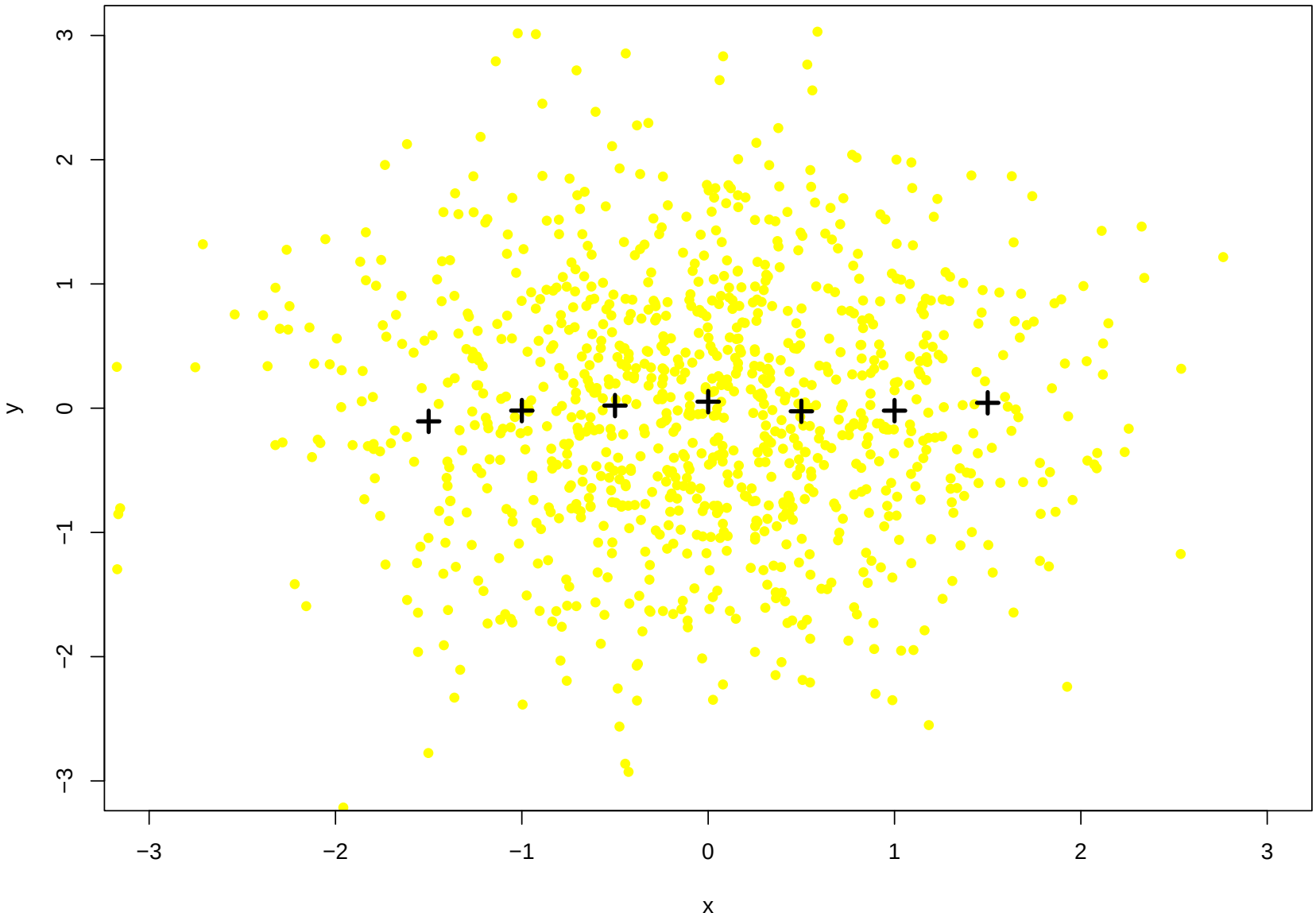
- Finish reading Chapters 13-14 and start reading Chapter 15 for the next lecture.
- Quiz 5 is next Wednesday in lecture

The Regression Effect and Regression Fallacy

- In virtually all test-retest situations, the bottom group on the first test will on average show some improvement on the second test, and the top group will on average fall back.
- Why?

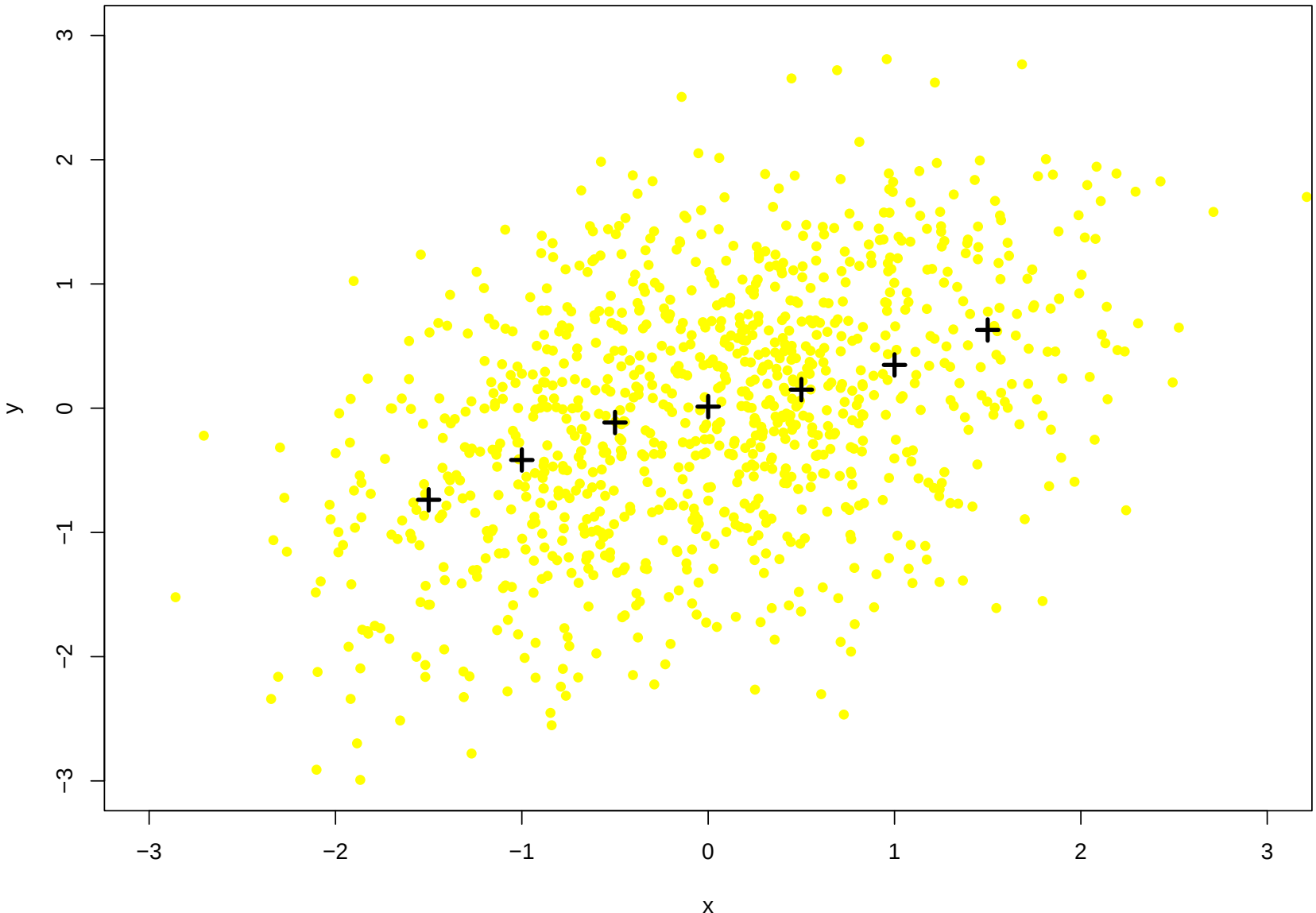
Thought Experiment

- Suppose one test is on math, the other on history.
- What would the scatter plot and graph of averages look like?



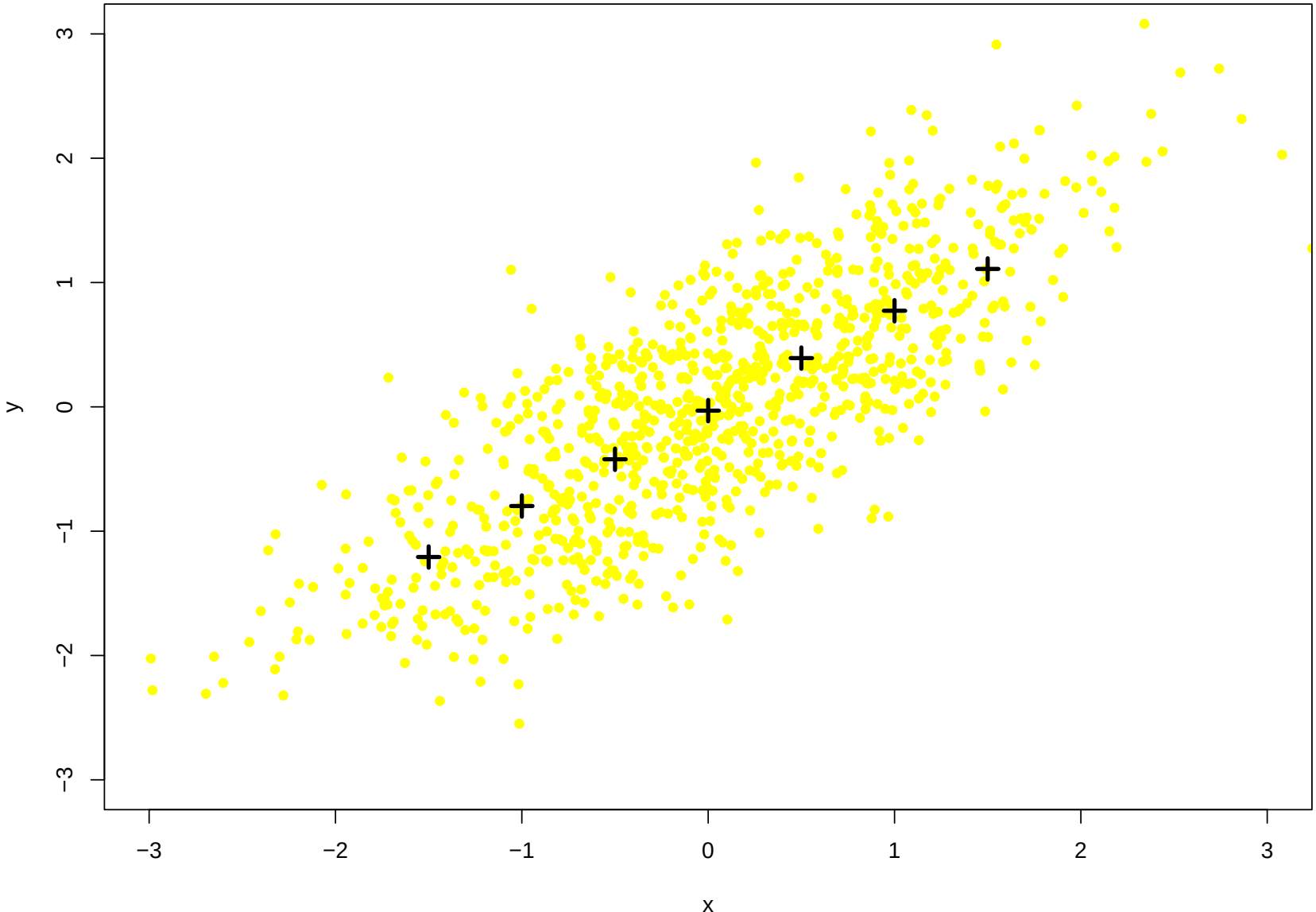
Thought Experiment Part 2

- Suppose one test is on math, the other on physics.
- What would the scatter plot and graph of averages look like



Thought Experiment Part 3

- Suppose the two tests are on the same topic.
- What would the scatter plot and graph of averages look like?



Regression Effect Conclusion

- In virtually all test-retest situations, the bottom group on the first test will on average show some improvement on the second test, and the top group will on average fall back.
- Why?

The Regression Fallacy

- It's easy to mistake the regression effect as something interesting.
- Example: Suppose a teacher takes the lowest 10 performing students from the midterm, and provides them with extra tutoring.
- On the final, their scores improve on average.
- Does this indicate the tutoring helped?
  - Not necessarily. It could very easily be nothing more than the regression effect.
  - How could you design a study to account for the regression effect?

Using Regression

The Regression Method

- **If the data follow a linear trend**, then:
- To estimate the average value of y for a given value of x without calculating the equation for the regression line,
  1. Convert x to standard units.
  2. Multiply by r. This gives you the average value of y in standard units.
  3. Convert from standard units back to the original scale for y.
- The same procedure works for percentiles:
  - Convert to and from standard units using the normal table.

Example 1

- In a certain class, midterm scores average out to 60 with an SD of 15, as do the scores for the final.
- The correlation between midterm scores and final scores is about 0.5.
- The data are football shaped.
- Predict the final score for a student whose midterm score was:
  - 75
  - 30
  - 60
  - unknown

Example 1 (continued)

- What is the root mean squared error for predicting the scores on the final based on the scores on the midterm?
  - How does this compare to the error for predicting the score on the final without using the score on the midterm?
- If the midterm score was 60, then what range would we expect 95% of the final scores to be in?

Example 2

- For the first-year students at a certain university, the correlation between SAT scores and first-year GPA was 0.7.
- The scatter diagram is football-shaped.
- Predict the percentile rank on the first-year GPA for a student whose percentile rank on the SAT was
  - 90%
  - 30%
  - 50%
  - unknown

Solutions to Example 1

1. $z = (75 - 60)/15 = 1, z_{final} = 1 * 0.5 = 0.5, pred = 60 + 0.5 * 15 = 67.5$
2. $z = (30 - 60)/15 = -2, z_{final} = -2 * 0.5 = -1, pred = 60 - 1 * *15 = 45$
3. If they score at the mean on the midterm, you would predict that they would score at the mean, or 60, on the final.
4. If you do not know the score on the midterm, then you would predict the mean score for the final, or 60.
5. The RMS error is $\sqrt{1 - 0.5^2} * 15 = 13$.
6. $60 \pm 2 * 13 = 34 - 86$.

Solutions to Example 2

1. The 90th percentile corresponds to 80% of the area in the middle, and the table in the book gives $z = 1.28$. The predicted z-score on the final would be $0.5 * 1.28 = 0.64$, and the percentile ranking is roughly $48\% + (100\% - 48\%)/2 = 74\%$.
2. The 30th percentile corresponds to 40% of the area in the middle, and the table in the book gives $z = -0.52$. The predicted z-score on the final would be $0.5 * -0.52 = -0.26$, and the percentile ranking is roughly $(100\% - 20\%)/2 = 40\%$.
3. The 50th percentile is the mean, and so we would predict the mean, or 50th percentile, on the final.
4. We would predict the mean, or 50th percentile, on the final.

Review Exercises

- See Review_exercises8.pdf
- lecture07_code.R gives the code that I used to create the plots in the exercises and has functions for the following:
  - displaying scatter plots
  - displaying conditional and marginal distributions,
  - calculating and displaying conditional means,
  - displaying graphs of averages, and
  - displaying the regression line and graph of averages on the scatter plot.
- Feel free to use the code in lecture07_code.R to help with the review exercises.

Reading Summary - Chapter 13

Section 13.1: Introduction

- Chance (Probability) is surprisingly difficult to define in practical terms.
  - In this class we will use the frequency theory.
  - This theory only works for processes that can be repeated over and over, independently, and under the same conditions.
  - The chance of something is the percentage of times it is expected to happen when the basic process is repeated many times.
  - Chances are between 0 and 1 (0% and 100%)
  - The chance of something equals 1 minus the chance of the opposite thing.
  - Basic examples.

Section 13.2: Conditional Probabilities

- Basic example of conditional probability:
  - You deal two cards from a standard 52 card deck.
    - What is the chance the second card is the queen of hearts? Answer: $1/52$
  - You deal two cards from a standard 52 card deck.
    - What is the chance the second card is the queen of hearts, **given the first card is the seven of clubs**? Answer: $1/51$
- The first question deals with the **unconditional** probability.
- The second question deals with the **conditional** probability.

Section 13.3: The Multiplication Rule

- The chance that two events will both happen equals the chance that the first will happen, multiplied by the chance the second will happen given the first has happened.
  - Some examples

Section 13.4: Independence

- Two events are **independent** if the chances for the second event given the first event are the same, no matter how the first one turns out.
  - Otherwise, the two events are **dependent**.
- When drawing at random with replacement, the draws are independent.
- When drawing at random without replacement, the draws are dependent.
- If two events are independent, the chance that both will happen is the product of their unconditional probabilities.
  - This is a special case of the multiplication rule.

Section 13.5: The Collins Case

- Interesting example from a court case
- Don't blindly multiply unconditional probabilities. Check for independence.
- Similar issues arise with DNA evidence.

Reading Summary – Chapter 14

Section 14.1: Listing the Ways

- One way to go about finding probabilities is to write down all of the possible outcomes of the chance process.
  - A critical question: are all the outcomes equally likely?

Section 14.2: The Addition Rule

- Two events are **mutually exclusive** when the occurrence of one prevents the occurrence of the other.
- If two events are mutually exclusive, the chance that at least one of them will occur is the sum the individual chances.
- If two events are not mutually exclusive, don't just add the chances because the result will be too big.

Section 14.3: Two FAQs

- Both independent and mutually exclusive can be roughly translated into ordinary English as meaning that the two events have nothing to do with each other.
- However, independent and mutually exclusive have very different meanings. **Watch out!**

Section 14.4: The Paradox of Chevalier De Mere

- Historical example about the chance that at double ones will turn up in a sequence of die rolls.
- A common, useful trick: if the chance of an event is hard to find, try to find the chance of the opposite event instead and subtract from 1.

Section 14.5: Are Real Dice Fair?

- To a good approximation, yes.
- This is an empirical fact, not a mathematical fact.

Basic Concepts and Notation

Frequency Interpretation of Probability

- Consider a sequence of identical trials, each of which may or may not result in an event A occurring.
  - Example: Flipping a coin repeatedly. One possible event would be that the coin lands heads.
- The **probability** of event A is the long-run fraction of times A occurs in many repetitions.

Probability: Frequency Interpretation vs Subjective Interpretation

- Statement: The probability of a coin landing heads is $1/2$.
- Frequency interpretation:
  - When a coin is flipped over and over, the long run percentage of heads is $1/2$.
- Subjective interpretation:
  - When a coin is flipped, I think it is just as likely to land heads as tails.
- There is a deep philosophical difference between these interpretations.
- What are conceptual problems / limitations with each of the interpretations?

A Common Formula for Probability

$$P(A) = \frac{\text{Number of ways } A \text{ can happen}}{\text{Total number of possible outcomes}}$$

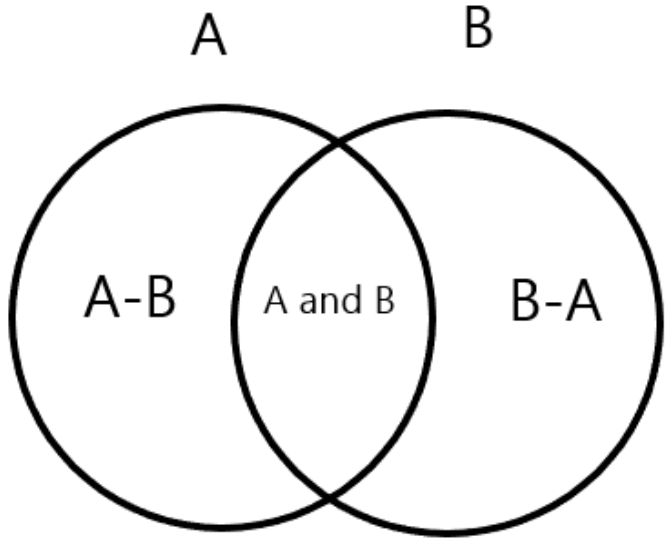
- Assumes each possible outcome is equally likely.
  - Not always obvious!
- This is **not** the definition of probability.

A Classic Example

- Suppose that you roll two six-sided dice, with numbers $1-6$ on each die, and take the sum of the numbers on the dice.
- What is the chance that the sum will be 2?
  - One approach would say that there are 11 possible outcomes $2-12$, and 2 is one of these outcomes.
  - Hence the probability is $1/11$.
  - Is this correct? Why or why not?

Visualizing Probability

- You can visualize probabilities using a Venn diagram.
- The probability is analogous to the area in the diagram.



Conditional Probability

- Probability of A given B

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

- This represents the proportion of B in the Venn diagram that is also in A.

Discussion Question

- Suppose you deal two cards from a standard 52 card deck.
- Define events $A =$ "First card is a King" and $B =$ "Second card is an Ace"
- What is $P(A)$?
- What is $P(B)$?
- What is $P(B|A)$?
- Explain in words the frequency interpretation for each of the above.
  - Note that the frequency interpretation of $P(B|A)$ does not involve the word "know."
- Compare this to the subjective interpretation.

Independence

- Events A and B are **independent** if and only if $P(A|B) = P(A)$ and $P(B|A) = P(B)$.
  - Does $P(A|B) = P(A)$ imply that $P(B|A) = P(B)$?
  - We could also write the definition as $P(A \text{ and } B) = P(A)P(B)$.
- In other words, the probability of A is the same whether or not B happens, and vice versa.

Examples of Independence

- Repeated tosses of a coin.
- Repeated draws of a card from a standard 52 card deck.
  - You put back (replace) the card after each draw.
  - Would the repeated draws be independent if you didn't replace the card?
- Voting decisions of individuals in different states who don't know each other.
- Other examples?

Mutually Exclusivity

- Events A and B are **mutually exclusive** if and only if $P(A \text{ and } B) = 0$.
- In other words, it is impossible for both A and B to happen.

Examples of Mutual Exclusivity

- Getting an A on an exam and getting a B on the exam.
- Winning and losing a game.
- Other examples?

Watch Out!!

- Both independent and mutually exclusive can be translated into ordinary English as meaning that the two events have nothing to do with each.
- These are very different concepts!
- Suppose that A and B are both independent and mutually exclusive.
  - What does this say about $P(A)$ or $P(B)$?

Calculation Rules

- It is always true that $P(A \text{ and } B) = P(A)P(B|A)$.
- A and B independent $\rightarrow P(A \text{ and } B) = P(A)P(B)$.
- If A and B mutually exclusive, then
  - $P(A \text{ or } B) = P(A) + P(B)$.
- How can we use this fact to show that
  - $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$?
- If A and B are independent, then
  - $P(A \text{ or } B) = P(A) + P(B) - P(A)P(B)$
- We can use these rules to put bounds on the **joint probability** $P(A \text{ and } B)$.

Bounds on Joint Probabilities

Joint Probabilities (Example 1)

- Suppose $P(A) = 0.4$ and $P(B) = 0.5$
  - What is $P(A \text{ and } B)$?
  - What is $P(A \text{ and } B)$ if A and B are independent?
  - What is $P(A \text{ and } B)$ if A and B are mutually exclusive?
  - Are there any other possible values that $P(A \text{ and } B)$ could be?
  - Are there any values that $P(A \text{ and } B)$ ***can't*** be?

Joint Probabilities (Example 2)

- Suppose $P(A) = 0.8$ and $P(B) = 0.5$
  - What is $P(A \text{ and } B)$?
  - What is $P(A \text{ and } B)$ if A and B are independent?
  - What is $P(A \text{ and } B)$ if A and B are mutually exclusive?
  - Are there any other possible values that $P(A \text{ and } B)$ could be?
  - Are there any values that $P(A \text{ and } B)$ ***can't*** be?

Bounds on Joint Probabilities (discussion)

- For any pair of events A and B,
  - What is the maximum possible value of $P(A \text{ and } B)$?
  - What is the minimum possible value of $P(A \text{ and } B)$?
  - What is the maximum possible value of $P(A \text{ or } B)$?
  - What is the minimum possible value of $P(A \text{ or } B)$?

Bounds on Joint Probabilities

- For any pair of events A and B,
  - The maximum possible value of $P(A \text{ and } B)$ is $\min(P(A), P(B))$.
  - The minimum possible value of $P(A \text{ and } B)$ is $\max(P(A) + P(B) - 1, 0)$.
  - The maximum possible value of $P(A \text{ or } B)$ is $\min(P(A) + P(B), 1)$.
  - The minimum possible value of $P(A \text{ or } B)$ is $\max(P(A), P(B))$.

Discussion Exercise

- If A is "rain on Saturday," and B is "rain on Sunday," and $P(A) = 0.4$ and $P(B) = 0.4$,
  - What is the maximum possible value of the probability of rain on both Saturday and Sunday?
  - What is the minimum possible value of the probability of rain on both Saturday and Sunday?

Review Questions

- See Review_questions9-10.pdf on Canvas